

second because of network congestion. 802.11n, however, reportedly can achieve speeds as high as 140 megabits per second. The standard is currently in draft form and is rumoured to be due for release by 2010.

Once the signals can be received and sent using a similar protocol, a wireless network can be established between two Wi-Fi enabled devices.

Wireless local area networks

WLAN is a wireless local area network that, as mentioned above, links two or more computers or devices using spread spectrum or OFDM modulation technology, in a limited area. This gives users the mobility to move around within a broad coverage area and still be connected to the network.

Wireless LAN technology was developed as early as 1991 when wireless LAN products had just appeared in the market and the IEEE 802.11 committee was just standardising the

Wi-Fi and wireless LAN have become more accessible and convenient in recent times due to falling equipment prices

protocols. In less than five years, the technology gained widespread application and implementation, with hospitals, colleges and stock exchanges all being enabled for nomadic access, point-to-point local area network bridges, ad-hoc networking and internet access.

We have discussed how all it takes to enable Wi-Fi technology is the small radio. However, initially, the WLAN hardware was so expensive it was used only as an alternative to cabled LAN in places where cabling was difficult or inefficient. However, since the IEEE standards emerged and there was a proper industry focus on interoperability and the growth of the Wi-Fi Alliance, the WLAN has taken great strides in development.

All components that can connect into a wireless medium in a network are referred to as stations. All stations are equipped with wireless network interface cards (WNICs) which contain the main hardware and embedded protocols for transmission of data across the spectrum.

Wireless stations fall into one of two categories: access points and clients. The basic service set (BSS) is a set of all